

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457258.833 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.022 +/- 0.061
Transit depth (Rp/Rs)^2	0.047 +/- 0.26 [%]
Orbital Inclination (inc)	83.54 +/- 1.9
Airmass coefficient 1 (a1)	0.074 +/- 0.038
Airmass coefficient 2 (a2)	2.42 +/- 1.55
Scatter in the residuals of the lightcurve fit is	46.73 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.014 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.022 ± 0.061 vs 0.1592 (deviation 1.87σ)
Duration fit vs published	0.014 d vs 0.0946 d (deviation 2.79σ)
Mid-time O-C	-5.7 min
Depth SNR	0.3
Residual scatter	46.73 %

Light curve

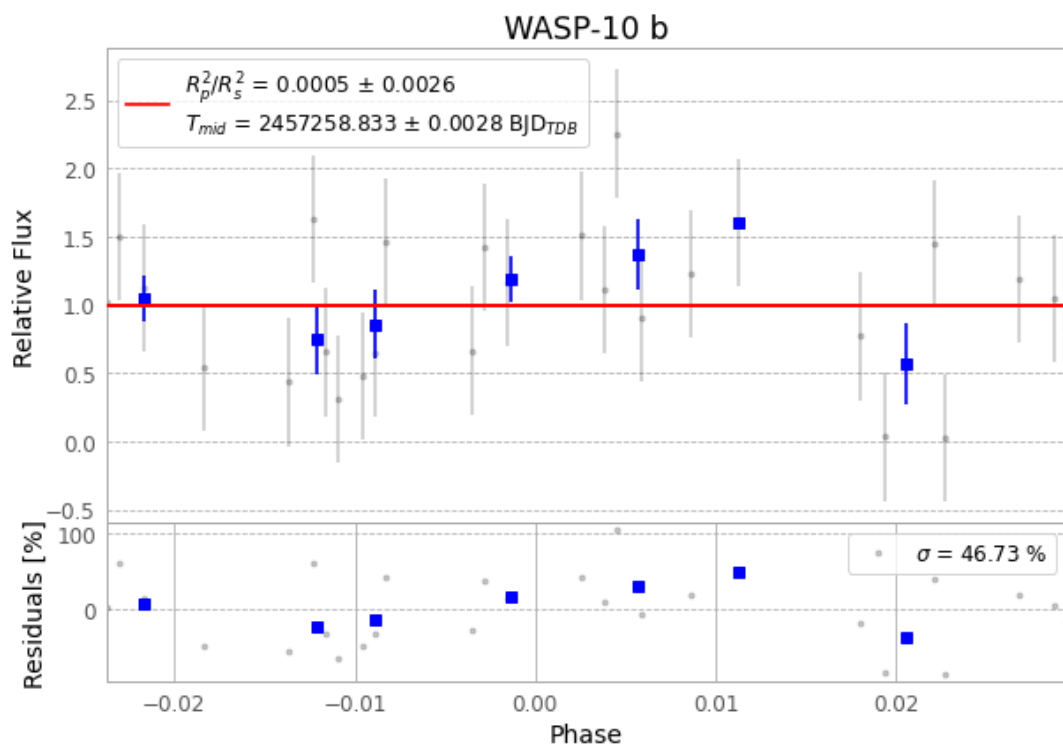


Figure 1 — FinalLightCurve_WASP-10 b_2015-08-23.png (SHA-256 79af2f7453b0949a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 442], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 62400f687d8f9615...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), WASP-10150824060012.FITS → WASP-10150824100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150824.log (f453efe27a9a...), FinalLightCurve_WASP-10 b_2015-08-23.png (79af2f7453b0...), FinalLightCurve_WASP-10 b_2015-08-23.csv (d9fbfbfd8dd28...)

Compute resources

Wall time 167.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:11Z.